



JKBOSE 2026

CLASS 11TH

CONIC SECTIONS

MATHS

LECTURE-5

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TOPICS TO BE COVERED

1

Ellipse

2

Related Problems

Ellipse

The locus of a point P in a plane such that the sum of its distances from two fixed points is constant, is known as an ellipse.

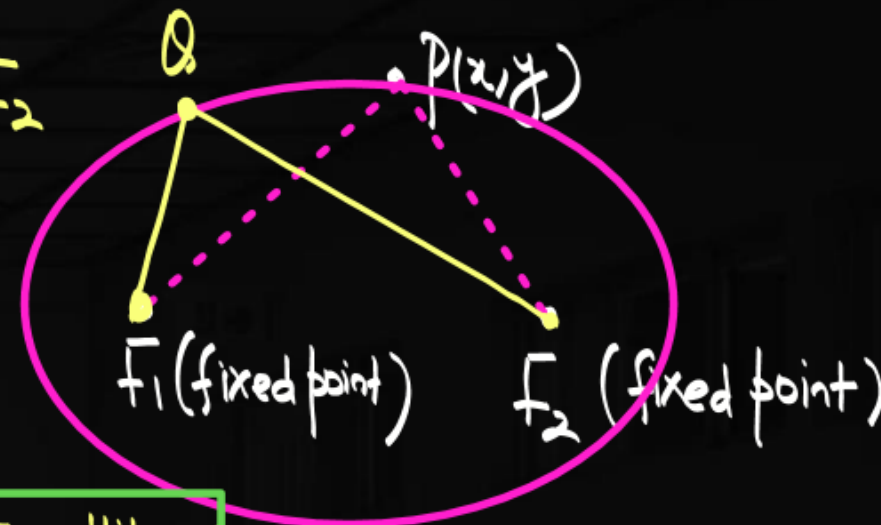
OR

The collection of points in a plane whose sum of the distances from two fixed points is always constant, is known as an ellipse.

$$\text{i.e. } PF_1 + PF_2 = \text{Constant}$$

* The fixed points are known as the foci of the ellipse.

$$QF_1 + QF_2 = PF_1 + PF_2 = \text{Constant}$$



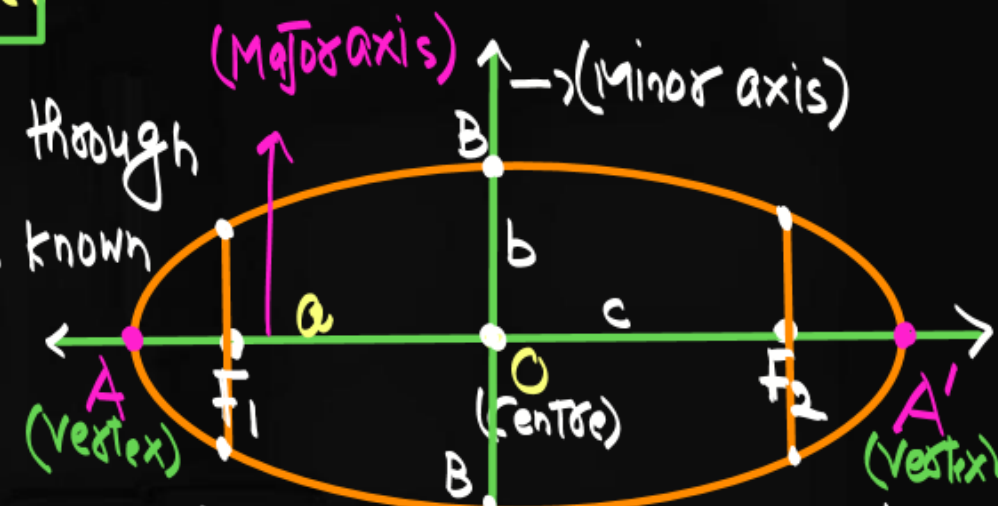
★ Terms Related To ellipse:

Major axis: Line passing through the foci of the ellipse is known as the major axis.

Vertices of an ellipse:

The points of intersection of the major axis & the ellipse is known as the vertices of an ellipse.

Centre of an ellipse: The mid-point of the line segment joining the two foci is known as the centre of the ellipse.



Minor axis: Line passing through the centre & is perpendicular to the major axis is known as the Minor axis.

Latus Rectum: Chord passing through the focus and is perpendicular to the major axis is known as the latus Rectum.

- * a $\longrightarrow$ Distance of one of the vertex from the centre.
- * b $\longrightarrow$ Distance of one of the point of intersections of ellipse & minor axis from the centre.
- * c $\longrightarrow$ Distance of one of the foci from the centre.
- * $2a$ $\longrightarrow$ Distance b/w the vertices of an ellipse.
(length of Major axis)

- * $2b$ $\longrightarrow$ Distance b/w B and B'
(length of Minor axis)
- * $2c$ $\longrightarrow$ Distance b/w the two foci

★ Relationship between a, b & c :

By Pythagoras Theorem

$$BF_2^2 = OF_2^2 + OB^2$$

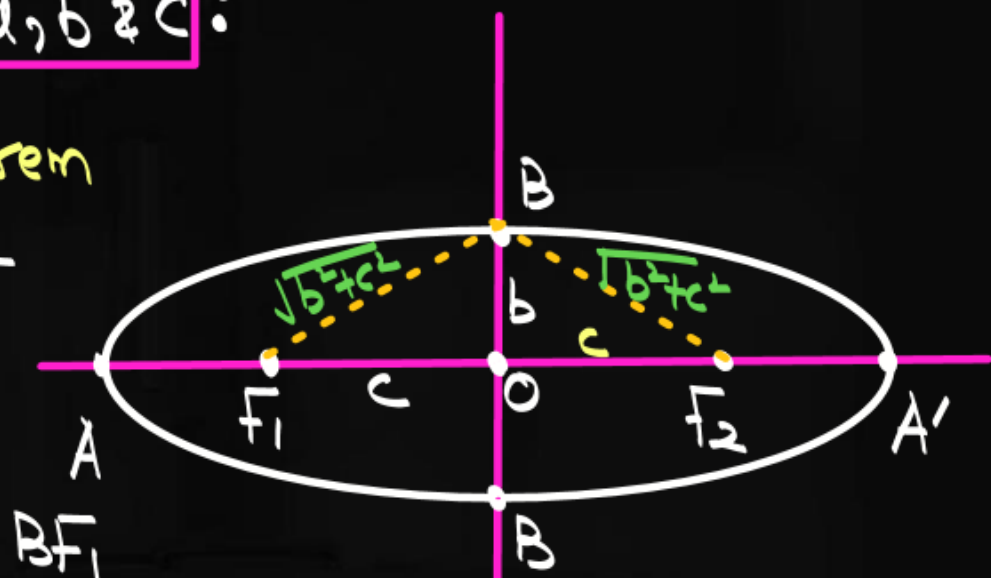
$$\Rightarrow BF_2^2 = c^2 + b^2$$

$$\Rightarrow BF_2 = \sqrt{c^2 + b^2} = BF_1$$

$$\Rightarrow \boxed{BF_1 + BF_2 = 2\sqrt{b^2 + c^2}} \quad \text{--- (1)}$$

Also $A'F_1 + A'F_2 = \cancel{\varphi} + a + a - \cancel{\varphi}$

$$\Rightarrow \boxed{A'F_1 + A'F_2 = 2a} \quad \text{--- (2)}$$



By the definition of an ellipse,

$$BF_1 + BF_2 = A'F_1 + A'F_2$$

$$\Rightarrow 2\sqrt{b^2 + c^2} = 2a \text{ (from (1) \& (2))}$$

$$\Rightarrow \sqrt{b^2 + c^2} = a$$

$$\Rightarrow b^2 + c^2 = a^2$$

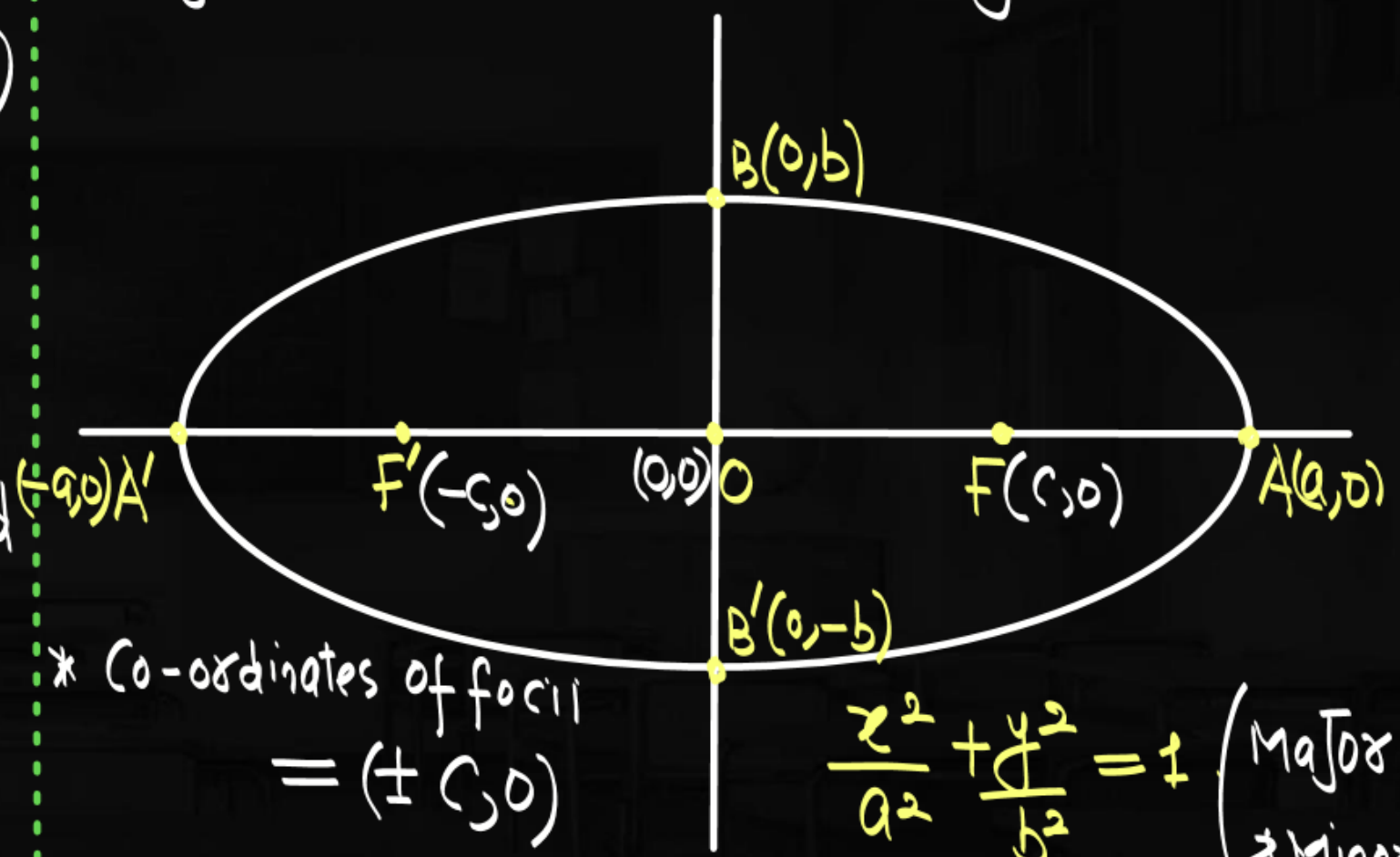
$$\Rightarrow c = \sqrt{a^2 - b^2} \text{, which is the required relation.}$$

eccentricity: The ratio of the distances from Centre of the ellipse to one of the foci & to one of vertices is known as eccentricity.

$$\Rightarrow e = \frac{c}{a} \left(\begin{array}{l} \text{clearly } c < a \\ \Rightarrow \frac{c}{a} < 1 \\ \Rightarrow e < 1 \end{array} \right)$$

★ Equation of the ellipse in the Standard form:

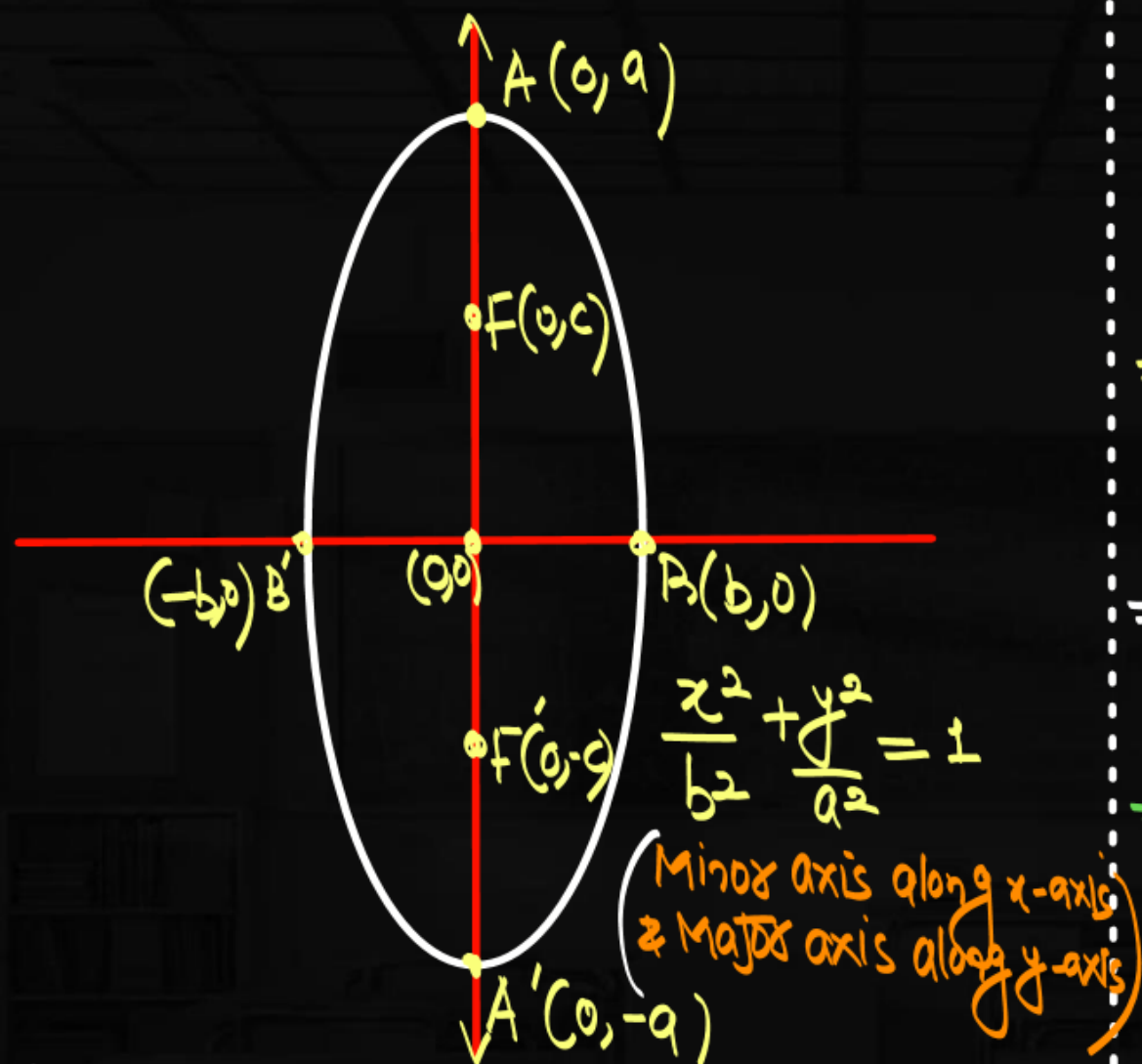
- * origin is considered to be the Centre of the ellipse.
- * Major axis & Minor axis are along the Co-ordinate axes.



* Co-ordinates of foci = $(\pm c, 0)$

* Co-ordinates of the vertices = $(\pm a, 0)$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \left(\begin{array}{l} \text{Major axis along } x\text{-axis} \\ \& \text{Minor axis along } y\text{-axis} \end{array} \right)$$



Co-ordinates of the foci = $(0, \pm c)$

Co-ordinates of the vertices
= $(0, \pm a)$.

Remark: we know that

$$e = \frac{c}{a}$$

$$\Rightarrow e^2 = \frac{c^2}{a^2} = \frac{a^2 - b^2}{a^2}$$

$$\Rightarrow e^2 = \frac{a^2}{a^2} - \frac{b^2}{a^2}$$

$$\Rightarrow e^2 = 1 - \frac{b^2}{a^2}$$

$$\Rightarrow \frac{b^2}{a^2} = 1 - e^2$$

$$\Rightarrow \boxed{b^2 = a^2(1 - e^2)}$$

Main Facts of Ellipse

Forms of the ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$
	$a > b$	$a > b$
Equation of major axis	$y = 0$	$x = 0$
Length of major axis	$2a$	$2a$
Equation of Minor axis	$x = 0$	$y = 0$
Length of Minor axis	$2b$	$2b$
Directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{a}{e}$
Equation of latus rectum	$x = \pm ae$	$y = \pm ae$
Length of latus rectum	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Centre	$(0, 0)$	$(0, 0)$

Thank
You